Workshop 6: General equilibrium

FIE463: Numerical Methods in Macroeconomics and Finance using Python

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See GitHub repository for notebooks and data:

https://github.com/richardfoltyn/FIE463-V26

Exercise 1: Labor supply without capital

Recall the consumption & labor choice problem studied in the lecture. In this exercise, we revisit this setting but assume that there is no capital in the economy.

Household problem

Households choose c and h to maximize utility

$$u(c,h) = \frac{c^{1-\gamma}-1}{1-\gamma} - \psi \frac{h^{1+1/\theta}}{1+1/\theta}$$

subject to the budget constraint

$$c = w \cdot h + \pi$$

where π are firm profits which are distributed to all households equally. Since all households are identical, we assume that the economy is populated by a *single* representative household.

Firm problem

We assume that firms have the decreasing-returns-to-scale production function

$$Y = zL^{1-\alpha}$$

where z is productivity (TFP) and labor L is the only input factor. Firms maximize profits Π,

$$\max_L \Pi = zL^{1-\alpha} - wL$$

which gives rise to the first-order condition

$$\frac{\partial \Pi}{\partial L} = (1-\alpha)zL^{-\alpha} - w = 0$$

We can solve for L to obtain the firm's optimal labor demand for given w:

$$L = \left(\frac{(1-\alpha)z}{w}\right)^{\frac{1}{\alpha}} \tag{1.1}$$

For simplicity, we assume there is a *single* firm which takes wages and the price of output as given, where the latter is normalized to one.

Equilibrium

The general equilibrium in this economy is a set of quantities (L, Y, Π, c, h, π) and the wage rate w which solve the household's and firm's problem, and the following conditions are satisfied:

- Labor market: $L = h$ (labor L demanded by firms equals hours h supplied by households).
- Goods market: $Y = c$ (aggregate output Y equals the amount of goods c consumed by households).
- Profits: $\Pi = \pi$ (profits distributed by firms equal profits received by households).

Analytical solution

By combining the household and firm first-order conditions, the problem can be reduced to a single equation in a single unknown, L (or h):

$$h = L = \left(\frac{(1-\alpha) z^{1-\gamma}}{\psi} \right)^{\frac{1}{1/\theta + \alpha + \gamma(1-\alpha)}} \tag{1.2}$$

We will use this expression later to compare the numerical to this exact solution.

Numerical solution

In the following, you are asked to adapt the code from the lecture to solve this problem. You should use the template file `workshop06_ex1.py` provided for this exercise to implement your solution.

1. Adapt the `Parameters` data class

   ```
   @dataclass
   class Parameters:
       pass
   ```

   so that it contains the following parameters as attributes: $\alpha = 0.36$, $z = 1$, $\gamma = 2$, $\psi = 1$, $\theta = 0.5$.

2. Implement the function `solve_hh(w, pi, par)` to solve the household problem for a given w and π. This function should return the household choices, in particular the **labor supply** h.

   Use the utility function `util(c, h, par)` defined in the template file for this purpose (this is the same function we used in the lecture).

3. Implement the function `solve_firm(w, par)` which returns the firm's **labor demand** L given by (1.1), output Y, and profits Π for a given wage w.

4. Implement the function `compute_labor_ex_demand(w, par)` which returns the **excess labor demand** for a given wage w.

5. Implement the function `compute_equilibrium(par)` which uses a root finder to locate the equilibrium, computes the equilibrium quantities (L, Y, Π, c, h, π) and wage rate w and stores these using an instance of the `Equilibrium` data class defined in `workshop06_ex1.py`.

6. Compute the equilibrium using the function you just implemented and print the quantities and prices using `print_equilibrium()` implemented in `workshop06_ex1.py` (you don't need to write this function yourself).

7. Compare your numerical solution to the analytical solution for the equilibrium L returned by `compute_analytical_solution()` implemented in `workshop06_ex1.py`.

Note: Include the following cell magic to automatically reload any changes you make to the template file:

[1]:
```
%load_ext autoreload
%autoreload 2
```

Solution.

The complete solution is provided in the file `workshop06_ex1.py`. In what follows, some of the code blocks from that file are replicated below to discuss the solution.

Part 1 — Parameters class

The definition of the `Parameters` class is almost identical to the lecture:

[2]:
```
from dataclasses import dataclass

@dataclass
class Parameters:
    """
    Container to store the problem's parameters.
    """
    alpha: float = 0.36     # Capital share in production function
    z: float = 1.0          # TFP
    gamma: float = 2.0      # RRA in utility
    psi: float = 1.0        # Weight on disutility of working
    theta: float = 0.5      # Frisch elasticity of labor supply
```

Part 2 — Household problem

The function to solve the household problem has to be adapted from the lecture to take into account the new budget constraint.

[3]:
```
from scipy.optimize import minimize
# Import the utility function implementation from the solution file
from workshop06_ex1 import util

def solve_hh(w, pi, par: Parameters):
    """
    Solve household problem for given prices and parameters.

    Parameters
    ----------
    w : float
        Wage rate
    pi : float
        Firm profits distributed to households
    par : Parameters
        Parameter instance

    Returns
    -------
    c_opt : float
        Optimal consumption
    h_opt : float
        Optimal labor supply
    """

    # Initial guess for h
```

```
    h_guess = 0.5

    res = minimize(
        lambda h: -util(w * h + pi, h, par),
        x0=h_guess,
        method='L-BFGS-B',
        bounds=((0, None), )
    )

    if not res.success:
        # Print diagnostic error message if minimizer had problems
        print('Minimizer did not terminate successfully')
        print(res.message)
        print(f'  Arguments: w={w}, pi={pi}')

    # Store optimal hours choice
    h_opt = res.x[0]
    # Optimal consumption follows from budget constraint
    c_opt = w * h_opt + pi

    return c_opt, h_opt
```

Part 3 — Firm problem

The function `solve_firm()` has to be adapted to reflect the fact that labor is now the only production factor:

[4]:
```
def solve_firm(w, par: Parameters):
    """
    Compute labor demand and profits implied by firm's first-order condition
    for given prices w.

    Parameters
    ----------
    w : float
        Wage rate
    par : Parameters
        Parameter instance

    Returns
    -------
    L : float
        Labor demand
    Y : float
        Output
    Pi : float
        Profits
    """

    # Labor demand
    L = ((1-par.alpha) * par.z / w)**(1/par.alpha)

    # Output
    Y = par.z * L**(1-par.alpha)

    # Profits
    Pi = Y - w * L

    return L, Y, Pi
```

Part 4 — Excess labor demand

The function `compute_labor_ex_demand()` calls the functions defined above to solve the household and firm problems, and computes the excess demand for labor as the difference between the firm's demand and the household's labor supply.

[5]:
```
def compute_labor_ex_demand(w, par: Parameters):
    """
    Compute excess demand for labor.

    Parameters
    ----------
    w : float
        Wage rate
    par : Parameters
        Parameter instance

    Returns
    -------
    ex_demand : float
        Excess demand for labor
    """

    # Wage and profits implied by firm's first-order condition
    L, Y, Pi = solve_firm(w, par)

    # Optimal household choices for given prices
    c_opt, h_opt = solve_hh(w, Pi, par)

    # Excess demand for labor
    ex_demand = L - h_opt

    return ex_demand
```

Part 5 — Equilibrium computation

The equilibrium quantities and prices are stored in the `Equilibrium` data class defined as follows:

[6]:
```
@dataclass
class Equilibrium:
    """
    Container to store equilibrium allocations and prices.
    """
    par: Parameters = None
    c: float = None
    h: float = None
    w: float = None
    L: float = None
    Y: float = None
    Pi: float = None
```

The following function calls the scalar root finder with an initial bracket to locate the equilibrium wage:

[7]:
```
from scipy.optimize import root_scalar

def compute_equilibrium(par):
    """
    Compute the equilibrium for given parameters.

    Parameters
    ----------
```

```
    par : Parameters
        Parameter instance

    Returns
    -------
    eq : Equilibrium
        Equilibrium instance containing equilibrium values
    """

    # Define initial bracket for root finder
    bracket = (1.0e-3, 5)

    res = root_scalar(
        compute_labor_ex_demand, bracket=bracket, args=(par, )
    )

    if not res.converged:
        print('Equilibrium root finder did not terminate successfully')

    # Create instance of equilibrium class
    eq = Equilibrium(par=par, w=res.root)

    # Equilibrium wage, output and profits
    eq.L, eq.Y, eq.Pi = solve_firm(eq.w, par)

    # Equilibrium household choices
    eq.c, eq.h = solve_hh(eq.w, eq.Pi, par)

    return eq
```

Part 6 — Equilibrium results

We can now initialize a `Parameters` object, call `compute_equilibrium()` and print the results. We use the `print_equilibrium()` function defined in `workshop06_ex1.py` to perform the latter task.

[8]:
```
from workshop06_ex1 import print_equilibrium

# Get instance of default parameter values
par = Parameters()

# Solve for equilibrium
eq = compute_equilibrium(par)

# Print equilibrium quantities and prices
print_equilibrium(eq)
```

```
Equilibrium:
  Households:
    c = 0.92453
    h = 0.88461
  Firms:
    Y = 0.92453
    L = 0.88461
    Pi = 0.33283
  Prices:
    w = 0.66888
  Market clearing:
    Labor market: -7.30551e-10
    Goods market: 4.88652e-10
```

Part 7 — Analytical vs Numerical

Finally, the module `workshop06_ex1.py` also contains a function to compute the analytical solution. We can use it to verify that the numerical solution is correct.

[9]:
```
from workshop06_ex1 import compute_analytical_solution

# Compare to analytical solution
L = compute_analytical_solution(par)
print(f'Analytical solution: h = L = {L:.5f}')
```

```
Analytical solution: h = L = 0.88461
```

As you can see, the analytical solution is basically identical to the numerical one.

Exercise 2: Unequal distribution of profits

We now extend the setting from Exercise 1 and assume that a fraction of households solely live on their labor income (type 1), while profits are only distributed to a subset of households (type 2). We can think of these households as workers and entrepreneurs, respectively. We assume the economy is populated by N_1 households of type 1 and N_2 households of type 2.

Household problem

All households have identical preferences which are unchanged from the previous exercise, but their budget constraints differ. For type-1 households, it is given by

$$c_1 = w \cdot h_1$$

whereas for type-2 households it's

$$c_2 = w \cdot h_2 + \pi_2$$

The subscripts in c_1, c_2, h_1, h_2, and π_2 index the household type since different households will choose different levels of consumption and labor supply.

Firm problem

The firm problem remains unchanged from the previous exercise. For convenience, we repeat the central equations:

$$\begin{aligned} \text{Labor demand:} \quad & L = \left(\frac{(1-\alpha)z}{w}\right)^{\frac{1}{\alpha}} \\ \text{Output:} \quad & Y = zL^{1-\alpha} \\ \text{Profits:} \quad & \Pi = zL^{1-\alpha} - wL \end{aligned}$$

Equilibrium

The general equilibrium in this economy is a set of quantities $(L, Y, \Pi, c_1, c_2, h_1, h_2, \pi_2)$ and the wage rate w which solve the household's and firm's problem, and the following conditions are satisfied:

- Labor market clearing: $L = N_1 h_1 + N_2 h_2$ (labor L demanded by firms equals hours supplied by households).

- Goods market clearing: $Y = N_1 c_1 + N_2 c_2$ (aggregate output equals the amount of goods consumed by households).
- Profits: $\Pi = N_2 \pi_2$ (profits distributed by firms equal profits received by type-2 households).

Numerical solution

In the following, you are asked to adapt the code you wrote for exercise 1 to solve the modified problem. The new solution only requires changes at a few selected points to take into account the unequal distribution of profits. You should use the template file `workshop06_ex2.py` provided for this exercise.

1. Adapt the `Parameters` class to include the two new parameters `N1` and `N2` which represent the number of type-1 and type-2 households, respectively. Set $N_1 = 5$ and $N_2 = 5$.

   For the remaining parameters, use the same values as in exercise 1.

2. Write the function `compute_labor_ex_demand(w, par)` which returns the excess labor demand for given w. Use the function `solve_hh()` and `solve_firm()` you wrote for exercise 1 to solve this task.

   Hint: Don't copy the implementations for `solve_hh()` and `solve_firm()` but directly import them from the module which contains the solution for exercise 1:

   ```
   from workshop06_ex1 import solve_firm, solve_hh
   ```

3. Write the function `compute_equilibrium(par)` which uses a root finder to locate the equilibrium, computes the equilibrium quantities $(L, Y, \Pi, c_1, h_1, c_2, h_2, \pi_2)$ and the wage rate w, and stores these using an instance of the `Equilibrium` data class defined in `workshop06_ex2.py`.

4. Compute the equilibrium using the function you just implemented and print the quantities and prices using `print_equilibrium()` defined in `workshop06_ex2.py`.

   How does the unequal distribution of profits affect consumption and labor supply of type-1 vs type-2 households?

5. You are interested to see if and how the allocation and prices in the economy change as we vary the number of type-1 and type-2 households. Assume that there are a total of $N = N_1 + N_2 = 10$ households in the economy

   - Using the function `compute_equilibrium()` you wrote earlier, compute the equilibrium when N_1 takes on the integer values from $0, \ldots, 9$ and $N_2 = N - N_1$.
   - Create a graph with four panels (2×2) which shows the aggregates Y, L, Π, and w as a function of N_1.
   - Create a graph with three columns which shows (c_1, c_2) in the first, (h_1, h_2) in the second, and π_2 in the third column. Use different colors and line styles to distinguish household types and include a legend.

     What do you conclude about the effects of inequality on the equilibrium allocation and prices?

Advanced solution method: Using analytical results and root-finding

Unlike the previous exercise, this economy no longer has a closed-form solution for the equilibrium quantities. From the households' first-order conditions, we can derive that the equilibrium is characterized by the two non-linear equations

$$(wh_1)^{-\gamma} = \psi \frac{h_1^{1/\theta}}{w}$$
$$(wh_1 + \pi_2)^{-\gamma} = \psi \frac{h_2^{1/\theta}}{w}$$

Note that w and $\pi_2 = \frac{\Pi}{N_2}$ itself are functions of (h_1, h_2) via the labor market clearing $L = N_1 h_1 + N_2 h_2$:

$$w = (1-\alpha) z L^{-\alpha} = (1-\alpha) z \big(N_1 h_1 + N_2 h_2\big)^{-\alpha}$$
$$\Pi = \alpha z L^{1-\alpha} = \alpha z \big(N_1 h_1 + N_2 h_2\big)^{1-\alpha}$$

We can substitute these two equations into the non-linear equation system above and numerically find a solution (h_1, h_2) that satisfies these conditions.

1. Use the multivariate root finder `root()` from `scipy.optimize` with `method='hybr'` to solve the above equation system.

   To do this, you need to write a function which takes as argument a vector `x` which contains the values (h_1, h_2), and return the errors in the two first-order conditions, i.e., a vector that contains the left-hand minus the right-hand side for each of the two equations.

2. Make sure the results obtained from this approach are the same as in the main exercise.

Solution.

The solution is provided in the file `workshop06_ex2.py`. In what follows, some of the code blocks from that file are replicated below to discuss the solution.

Part 1 — Parameters class

We extend the definition of the `Parameters` from exercise 1 to include the additional parameters `N1` and `N2`:

[10]:
```
from dataclasses import dataclass


@dataclass
class Parameters:
    alpha: float = 0.36  # Capital share in production function
    z: float = 1.0  # TFP
    gamma: float = 2.0  # RRA in utility
    psi: float = 1.0  # Weight on disutility of working
    theta: float = 0.5  # Frisch elasticity of labor supply
    N1: int = 5  # Number of type-1 households
    N2: int = 5  # Number of type-2 households
```

Part 2 — Excess labor demand

The function `compute_labor_ex_demand()` needs to be adapted from exercise 1 since now profits are only distributed to type-2 households. We directly use the implementations of the household and firm problems from Exercise 1, and compute the excess demand for labor as the difference of the firm's demand and the household's labor supply.

Note that now the aggregate labor demand is the sum of labor supplies of type-1 and type-2 households.

[11]:
```python
# Import functions from exercise 1
from workshop06_ex1 import solve_firm, solve_hh


def compute_labor_ex_demand(w, par: Parameters):
    """
    Compute the excess demand for labor.

    Parameters
    ----------
    w : float
        Wage rate.
    par : Parameters
        Model parameters.

    Returns
    -------
    float
        Excess demand for labor.
    """
    # Wage and profits implied by firm's first-order condition
    L, Y, Pi = solve_firm(w, par)

    # Equilibrium household choices
    # Type 1 receives no profits
    c1, h1 = solve_hh(w, 0.0, par)
    # Per-capita profits for type 2 households
    pi2 = Pi / par.N2
    c2, h2 = solve_hh(w, pi2, par)

    # Excess demand for labor
    ex_demand = L - par.N1 * h1 - par.N2 * h2

    return ex_demand
```

Part 3 — Equilibrium computation

Lastly, the `compute_equilibrium()` function needs to be adapted from exercise 1 to take into account the unequal distribution of profits.

[12]:
```python
from scipy.optimize import root_scalar


def compute_equilibrium(par):
    """
    Compute the equilibrium of the model.

    Parameters
    ----------
    par : Parameters
        Model parameters.

    Returns
    -------
    Equilibrium
        Equilibrium object containing the equilibrium values.
    """
    # Initial bracket for root finder
    bracket = (1.0e-3, 5)

    # Find equilibrium wage
```

```
    res = root_scalar(compute_labor_ex_demand, bracket=bracket, args=(par,))

    if not res.converged:
        print('Equilibrium root finder did not terminate successfully')

    # Create instance of equilibrium class
    eq = Equilibrium(par=par, w=res.root)

    # Equilibrium wage, output and profits
    eq.L, eq.Y, eq.Pi = solve_firm(eq.w, par)

    # Equilibrium household choices
    eq.c1, eq.h1 = solve_hh(eq.w, 0.0, par)
    # Per-capita profits for type 2 households
    eq.pi2 = eq.Pi / par.N2
    eq.c2, eq.h2 = solve_hh(eq.w, eq.pi2, par)

    return eq
```

Part 4 — Equilibrium results

We can now initialize a `Parameters` object, call `compute_equilibrium()` and print the results using `print_equilibrium()` defined in `workshop06_ex2.py`.

[13]:
```
from workshop06_ex2 import print_equilibrium

# Get instance of default parameter values
par = Parameters()

# Solve for equilibrium
eq = compute_equilibrium(par)

# Print equilibrium quantities and prices
print_equilibrium(eq)
```

```
Equilibrium:
  Households:
    Type 1 (N = 5):
      c1 = 0.37079
      h1 = 1.39195
    Type 2 (N = 5):
      c2 = 0.57936
      h2 = 0.89085
      pi2 = 0.34205
  Firms:
    Y = 4.75073
    L = 11.41401
    Pi = 1.71026
  Prices:
    w = 0.26638
  Market clearing:
    Labor market: 4.10562e-09
    Goods market: -1.09365e-09
    Profits: 0.00000e+00
```

As you can see, type-1 households choose to work more and consume less compared to type-2 households. This is of course due to the fact that type-2 households receive extra income in the form of profits.

Part 5 — Comparative statics

We use the implementation from the `workshop06_ex2` module to compute the equilibria for each value of N_1. Within the loop body, we need to update the `N1` and `N2` attributes of the `Parameters` instance.

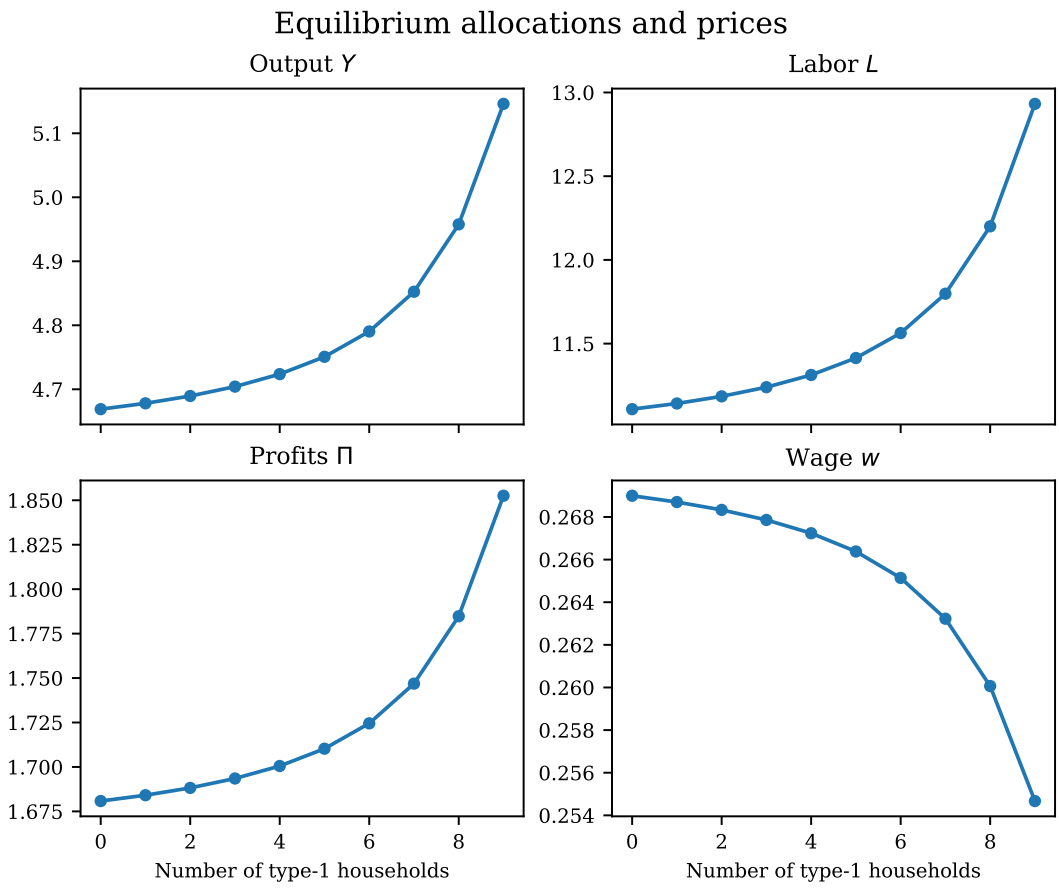
[14]:
```
import workshop06_ex2 as ex2
import numpy as np

N = 10
N1_grid = np.arange(0, N)
equilibria = []

for N1 in range(0, N):
    # Create new parameter instance with updated N1 and N2
    par = ex2.Parameters(N1=N1, N2=N - N1)

    # Compute equilibrium
    eq = ex2.compute_equilibrium(par)

    equilibria.append(eq)
```

Next, we pack the individual equilibrium quantities and the wage rate into individual lists.

[15]:
```
# Store equilibrium allocations and prices in separate lists
Y_grid = [eq.Y for eq in equilibria]
L_grid = [eq.L for eq in equilibria]
Pi_grid = [eq.Pi for eq in equilibria]
w_grid = [eq.w for eq in equilibria]
c1_grid = [eq.c1 for eq in equilibria]
c2_grid = [eq.c2 for eq in equilibria]
h1_grid = [eq.h1 for eq in equilibria]
h2_grid = [eq.h2 for eq in equilibria]
pi2_grid = [eq.pi2 for eq in equilibria]
```

We now create the graph showing the aggregates (Y, Π) and the wage rate w against the number of type-1 households in the economy.

[16]:
```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(
    2, 2, figsize=(6, 5), sharex=True, sharey=False, constrained_layout=True
)
ax[0, 0].plot(N1_grid, Y_grid, marker='o', ms=4)
ax[0, 0].set_title('Output $Y$')

ax[0, 1].plot(N1_grid, L_grid, marker='o', ms=4)
ax[0, 1].set_title('Labor $L$')

ax[1, 0].plot(N1_grid, Pi_grid, marker='o', ms=4)
ax[1, 0].set_title(r'Profits $\Pi$')
ax[1, 0].set_xlabel('Number of type-1 households')

ax[1, 1].plot(N1_grid, w_grid, marker='o', ms=4)
ax[1, 1].set_title('Wage $w$')
ax[1, 1].set_xlabel('Number of type-1 households')

_ = fig.suptitle('Equilibrium allocations and prices', fontsize=12)
```

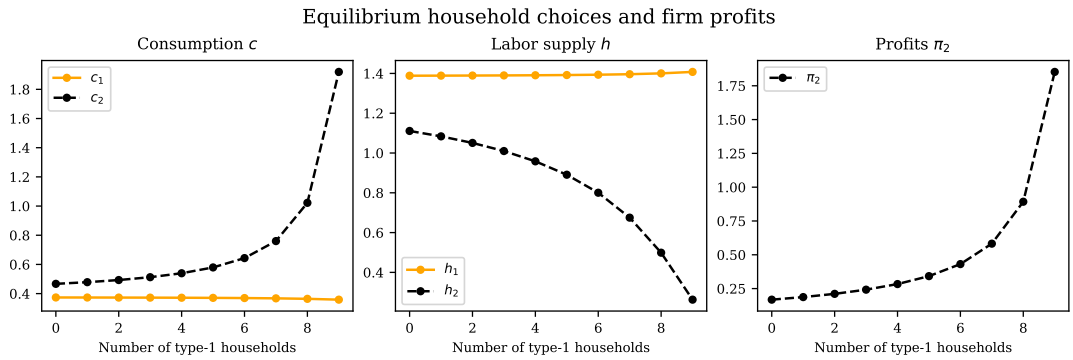


As can be seen, aggregate labor L and output increase in the number of type-1 households since these households supply more labor than type-2 households. At the same time, the wage rate drops slightly to accommodate this increase in labor supply. Lastly, profits are a fixed share of output, so they increase in tandem with output.

The next figure shows how household-level variables change in equilibrium as the number of type-1 households increases.

[17]:
```
fig, ax = plt.subplots(
    1, 3, figsize=(9, 3), sharex=True, sharey=False, constrained_layout=True
)

ax[0].plot(N1_grid, c1_grid, marker='o', ms=4, color='orange', label='$c_1$')
ax[0].plot(N1_grid, c2_grid, marker='o', ms=4, color='black', ls='--', label='$c_2$')
ax[0].set_title('Consumption $c$')
ax[0].legend()
ax[0].set_xlabel('Number of type-1 households')

ax[1].plot(N1_grid, h1_grid, marker='o', ms=4, color='orange', label='$h_1$')
ax[1].plot(N1_grid, h2_grid, marker='o', ms=4, color='black', ls='--', label='$h_2$')
ax[1].set_title('Labor supply $h$')
ax[1].legend()
ax[1].set_xlabel('Number of type-1 households')

ax[2].plot(
    N1_grid, pi2_grid, marker='o', ms=4, color='black', ls='--', label=r'$\pi_2$'
)
ax[2].set_title(r'Profits $\pi_2$')
ax[2].legend()
ax[2].set_xlabel('Number of type-1 households')
```

```
_ = fig.suptitle('Equilibrium household choices and firm profits', fontsize=12)
```



We see that for each N_1, type-1 households consume less and supply more labor than type-2 households. Their choices diverge at a fast rate for higher values of N_1 because profits are more and more concentrated among a smaller fraction of the population (and for $N_1 = 9$, only a single households receives all the profits).

Advanced solution method

Part 1 — Multivariate root finder

The alternative root-finding approach is implemented in the function `foc_error(x, par)`:

[18]:
```
def foc_error(x, par: Parameters):
    """
    Compute errors in first-order conditions of the household problem
    for type 1 and type 2.

    (for bonus question)

    Parameters
    ----------
    x : array_like
        Candidate guess for labor supply (h1, h2).
    par : Parameters
        Model parameters.

    Returns
    -------
    numpy.ndarray
        Array containing the differences from the first-order conditions.
    """
    # Extract candidate guess for labor supply
    h1, h2 = x

    # Aggregate labor supply
    L = par.N1 * h1 + par.N2 * h2
    # wage from firm's FOC
    w = (1 - par.alpha) * par.z * L ** (-par.alpha)
    # Aggregate firm profits
    Pi = par.alpha * par.z * L ** (1 - par.alpha)

    # FOC for HH type 1
    diff1 = par.psi * h1 ** (1 / par.theta) / w - (w * h1) ** (-par.gamma)
    # Profits per capita for HH type 2
    pi2 = Pi / par.N2
    # FOC for HH type 2
```

```
    diff2 = par.psi * h2 ** (1 / par.theta) / w - (w * h2 + pi2) ** (-par.gamma)

    fx = np.array((diff1, diff2))

    return fx
```

The function `foc_error()` can then be passed to the multivariate root finder as follows:

[19]:
```
from scipy.optimize import root

# Initial guess for labor supply (h1, h2)
x0 = np.array((0.5, 0.5))

# Find (h1, h2) that satisfy the FOCs
res = root(foc_error, x0=x0, args=(par,), method='hybr')
```

You can inspect the result object to confirm that the root finder terminated successfully. The root is stored in the vector `res.x` of the result object, and `res.fun` contains the multivariate function values at the root (which are numerically very close to zero).

[20]:
```
res
```

[20]:
```
 message: The solution converged.
 success: True
  status: 1
     fun: [-4.195e-09  1.139e-09]
       x: [ 1.408e+00  2.629e-01]
  method: hybr
    nfev: 17
    fjac: [[-9.968e-01 -7.966e-02]
           [ 7.966e-02 -9.968e-01]]
       r: [-1.647e+01  1.386e+00 -2.662e+00]
     qtf: [ 2.347e-08 -1.499e-08]
```

Part 2 — Verification

The solution in the module `workshop06_ex2` contains the function `compute_equilibrium_root()` which runs the root finder and computes the remaining equilibrium quantities. We can use this to compute and print the result:

[21]:
```
import workshop06_ex2 as ex2

# Compute equilibrium using root-finding approach
eq = ex2.compute_equilibrium_root(par)

# Print equilibrium values using print function we defined earlier
ex2.print_equilibrium(eq)
```

```
Equilibrium:
  Households:
    Type 1 (N = 9):
      c1 = 0.35850
      h1 = 1.40768
    Type 2 (N = 1):
      c2 = 1.91950
      h2 = 0.26291
      pi2 = 1.85255
  Firms:
    Y = 5.14597
    L = 12.93206
    Pi = 1.85255
  Prices:
```

```
    w = 0.25467
  Market clearing:
    Labor market: -8.88178e-16
    Goods market: 8.88178e-16
    Profits: 0.00000e+00
```

As you can see, the results are basically identical to those we computed earlier.